

THINGS TO KNOW:

1. Lab report must contain following sections: (order must be maintained)
 - a) Title /Question
 - b) Theory: The brief overview of the concept /techniques/syntax/technology used in the program
 - c) Code: The complete code (markups, styles and scripts)
 - d) Output: Screenshot of the output (in console and/or browser window)
2. Output screen should be captured (use snipping tool), printed and attached in the report. Other contents must be handwritten.
3. Every source code must include the printing statements to print following information after your main output:

Lab No.:

Name:

Roll No./Section :

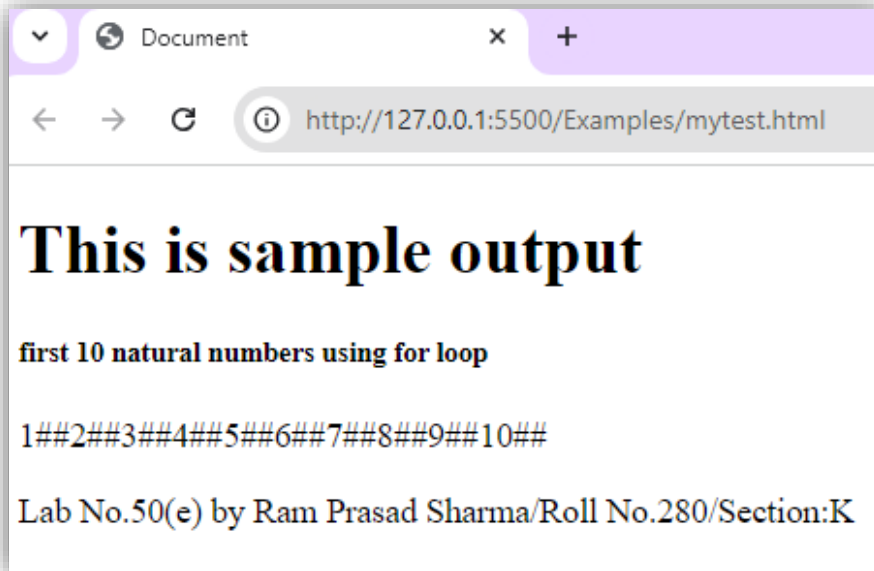
4. Contents should be written on single side of A4 sized paper.
5. The works must be submitted within specified deadline.
6. Cover page and index page should be attached in the report appropriately.

Index Page Format (can be printed)

List of Lab Works

Lab No.	Title /Question	Submission Date	Signature	Remarks
1(a)	This is sample	2079/11/12		
1(b)	This is also sample	2078/11/12		

Sample Output: (must be readable, showing browser address bar or console window menus)



[Lab reports after deadline will not be accepted. If you have any genuine reasons for not doing tasks in time inform the instructor before deadline!]

Scripting Language / BCA – 4th Semester Lab Works (Part-2)

12. Write a JS program to highlight all of the words over eight characters long in the paragraph text (with yellow background) when a button below the paragraph is clicked.
13. Create a webpage with following functionality.
 - ☞ There should be three radio buttons labeled with three colors. When one is selected the background of the page must be changed to respective color.
 - ☞ There should be an image, when mouse cursor is placed over the image, description of the image should be displayed on a paragraph below that image and when mouse cursor is removed from the image, the description should be vanished.
14. Working with HTML DOM
 - a) Create a HTML page and write a JS program to demonstrate the use of following properties of any node objects in DOM

nodeName	nodeType	lastChild
nodeValue	parentNode	nextSibling

previosSibling childNodes firstChild

b) Create a web page and write Script with JS for Selecting Document Elements as mentioned below:

- Selecting element by ID
- Selecting elements by name
- Selecting elements by tag type
- Selecting elements by CSS class
- Selecting element by querySelector()
- Selecting element by querySelectorAll()

c) Create a web page and write Script using JS for demonstrating

- Node creation
- Node insertion (using appenChild() and insertBefore())
- Node replacement (using replaceChild())
- Node Removal (using removeChild())

d) Demonstrate the use of following methods in DOM to work with attributes of HTML document

getAttribute() setAttribute()
hasAttribute() removeAttribute()

e) Create a form to enter the record of books. The interface should be as in following figure and descriptions:

Add Book Records

Book Id: Title: Author(s): Price:

Book id initially should be 100 and increments by 1 on every click of plus button. This value should be shown on textbox but cannot be editable by user. When plus button is clicked similar row should appear below it.

- f) Create a JS array containing some districts of your provenience. Now display those districts in unordered list dynamically on web page using DOM manipulation.

15. Understanding Regular Expression in JS

- a) Write a JS program to demonstrate the use of following concepts
- two ways of creating regular expression pattern in JS
 - using **exec()**, **test()**, **match()**, **matchAll()**, **search()**, **replace()**, **replaceAll()** and **split()** functions.
- b) Create and test for following Regex patterns in JS
- i. The first character of a string must be uppercase.
 - ii. A given value must contain a question marks, a dash and an underscore.
 - iii. The value must start from A or a and must end with full stop (.) or exclamation (!).
 - iv. The value must contain only one blank space.
 - v. The string must not contain any digits.
 - vi. The string must contain at least a symbol from the group of symbol (, , @, \$, %, !, &, *, ^, ~, -, +, (,), {, }, [,], ., :)
 - vii. The value must not contain any special character and digits.
 - viii. The value must contain @ symbol between two words each of at least one character.
 - ix. The string must not start with digit.
 - x. The string must start with underscore character or alphabets.
 - xi. The value must start with 'BCA-TU-' and after these character there must be only digits.
 - xii. The value must be in email address format.
 - xiii. The value must contain at least a vowel letter.
 - xiv. The value contain ten digits and first digit cannot be 0.
 - xv. The value should be email address and its must end with 'edu.np'
 - xvi. The value should be email address and it must contain 'gmail' or 'yahoo' after @ symbol.
 - xvii. The regex must match if the string contains 'nepal' case insensitively.

16. Forms and Client Side Validation using JS.

- a) Write a JS program to demonstrate the use of forms object (document.forms)
- b) Create a registration form and validate using separate JS file (should be from scratch. API /Library is not allowed). The form must include following entries and validation rules:
 - First Name (required , must contain alphabets only the length should be 1 to 50)
 - Last Name (optional, must contain alphabets only the length should be 1 to 50)
 - Date of birth (required, use date type)
 - Gender (required, must be selected from combo box)
 - Provenience (required, must be selected from combo box, 7 Provenience of Nepal)
 - District (required, must be selected from combo box, 77 districts of Nepal)
 - Cell Phone (required, 10 digits only not other symbols and alphabets)
 - Email address (optional, if entered must be in email address format)
 - User Name (required, must start with alphabet, can contain digits and only two special character (@ and _) are allowed, length must be in between 6 to 15 characters)
 - Password (required, must contain at least a digit, a special character, an uppercase letter and a lowercase letter, length must be in between 8 to 20 characters)
 - Confirm password (required, must match the value entered in password field)
 - Register (a submit button)

Show the error message with guideline next to the element in red color if the value is not satisfying the rule.

17. Create a web application using JS to simulate simple calculator which can perform addition, subtraction, multiplication and division for both integer and floating point numbers. Your application should contain all the required digits and symbols, displaying area and necessary other designs. The application should be able to response the input from keyboard press and mouse click both. [you can print the whole content for this lab task]
18. Cookies in JS

Write a JS program to demonstrate following concepts:

- Different ways of cookie creation
- Reading cookie
- Changing cookie
- Deleting cookie